4.8 Analyzing Graphs of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.8 (30-minute plan) · Student sheet

I can…

- use x-intercepts and end behavior to graph polynomial functions.
- find turning points and local maximums and minimums.
- tell whether a function is even, odd, or neither.

Vocabulary · fill in the missing word

Local maximum / minimum — the _____ of a turning point that is higher / lower than all nearby points

Turning points — a _____ of degree n has at most n − 1 turning points

Even function — f(−x) = f(x); _____ about the y-axis

Odd function — f(−x) = −f(x); _____ about the origin

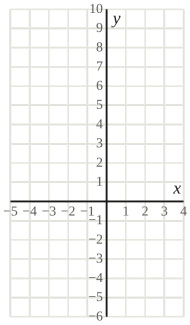
Multiplicity — the number of times a factor _____; even multiplicity → the graph touches the x-axis, odd → it crosses

Part 1 · Notes and examples (fill in each note, then work the example)

Multiplicity: Even multiplicity: the graph _____ the x-axis. Odd: it _____.

Example 1

Graph $f(x) = \frac{1}{2}(x + 3)(x - 2)^2$.



Turning points: A degree-n polynomial has at most _____ turning points.

Even and odd functions: Even function: symmetric about the _____. Odd function: symmetric about the _____.

Example 2

Find all real zeros of $f(x) = x^3 - x^2 - 6x$.

Part 2 · Partner practice

Partner practice

Find all real zeros of $f(x) = x^4 - 9x^2$.